LIST OF PUBLICATIONS

[**updated: July 14, 2026**]

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[60] M. Coti Zelati, M. Sorella & D. Villringer, *A fast dynamo on the three-torus,* submitted (2026). arXiv: 2603.09861

[59] M. Coti Zelati, L. Ertzbischoff & D. Gérard-Varet, *Stability analysis for active Brownian particle models,* submitted (2025). arXiv: 2512.17649

[58] M. Coti Zelati, M. Hairer & D. Villringer, *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise,* submitted (2025). arXiv: 2507.11422

[57] M. Coti Zelati, M. Sorella & D. Villringer, *Alpha-unstable flows and the fast dynamo problem,* submitted (2025). arXiv: 2504.00855

[56] M. Coti Zelati, A. Del Zotto & K. Widmayer, *On the stability of viscous three-dimensional rotating Couette flow,* submitted (2025). arXiv: 2501.17735

[55] M. Coti Zelati & V. Navarro-Fernández, *Three-dimensional exponential mixing and ideal kinematic dynamo with randomized ABC flows,* submitted (2024). arXiv: 2407.18028

[54] M. Coti Zelati & M.G. Delgadino, *Entropy maximization in the two-dimensional Euler equations,* Analysis & PDE **19** (2026), 505–538. DOI: 10.2140/apde.2026.19.505, arXiv: 2405.14738

[53] M. Coti Zelati, H. Dietert & D. Gérard-Varet, *Nonlinear stability for active suspensions,* Annales de l'Institut Henri Poincaré C. Analyse Non Linéaire **43** (2026), 1145–1199. DOI: 10.4171/AIHPC/164, arXiv: 2404.01906

[52] R.Bianchini, M. Coti Zelati & L. Ertzbischoff, *Ill-posedness of the hydrostatic Euler-Boussinesq equations and failure of hydrostatic limit,* Communications in Mathematical Physics **406** (2025), Art. 254. DOI: 10.1007/s00220-024-05224-y, arXiv: 2403.17857

[51] M. Coti Zelati, A. Del Zotto & K. Widmayer, *Stability of viscous three-dimensional stratified Couette flow via dispersion and mixing,* submitted (2024). arXiv: 2402.15312

[50] G.-Q. Chen, M. Coti Zelati & C.C. Yeung, *Vanishing Mach number limit of stochastic compressible flows,* submitted (2023). arXiv: 2311.14660

[49] M. Coti Zelati, T.D. Drivas, & R.S. Gvalani, *Statistically self-similar mixing by Gaussian random fields,* Journal of Statistical Physics **191** (2024), Art. 61. DOI: 10.1007/s10955-024-03277-w, arXiv: 2309.15744

[48] R. Bianchini, M. Coti Zelati & M. Dolce, *Symmetrization and asymptotic stability in non-homogeneous fluids around stratified shear flows,* Séminaire Laurent Schwartz – EDP et applications **5** (2022-2023), Art. 20, 17pp. DOI: 10.5802/slsedp.160, arXiv: 2309.12738

[47] M. Coti Zelati & M. Nualart, *Limiting absorption principles and linear inviscid damping in the Euler-Boussinesq system in the periodic channel,* Communications in Mathematical Physics **406** (2025), Art. 57. DOI: 10.1007/s00220-024-05224-y, arXiv: 2309.08445

[46] M. Coti Zelati & M. Nualart, *Explicit solutions and linear inviscid damping in the Euler-Boussinesq equation near a stratified Couette flow in the periodic strip,* Journal of Hyperbolic Differential Equations **21** (2024), 659-680. DOI: 10.1142/S0219891624400058, arXiv: 2309.08419

[45] M. Coti Zelati & A. Del Zotto, *Suppression of lift-up effect in the 3D Boussinesq equations around a stably stratified Couette flow,* Quarterly of Applied Mathematics **83** (2025), 389-401.(2025) DOI: 10.1090/qam/1697, arXiv: 2309.06426

[44] M. Coti Zelati, G. Crippa, G. Iyer & A.L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay,* Notices of the American Mathematical Society **71** (2024), 593–604. DOI: 10.1090/noti2929, arXiv: 2308.00358

[43] M. Coti Zelati, M. Dolce & C. Lo, *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes,* Journal of Mathematical Fluid Mechanics **26** (2024), Art. 12. DOI: 10.1007/s00021-023-00845-0, arXiv: 2305.18162

[42] E. Bruè, M. Coti Zelati & E. Marconi, *Enhanced dissipation for two-dimensional Hamiltonian flows,* Archive for Rational Mechanics and Analysis **248** (2024), Art. 84. DOI: 10.1007/s00205-024-02034-3, arXiv: 2211.14057

[41] J. Bedrossian, M. Coti Zelati & M. Dolce, *Taylor dispersion and phase mixing in the non-cutoff Boltzmann equation on the whole space,* Proceedings of the London Mathematical Society **129** (2024), Paper No. e12616, 70pp. DOI: 10.1112/plms.12616, arXiv: 2211.05079

[40] M. Coti Zelati, H. Dietert & D. Gérard-Varet, *Orientation mixing in active suspensions,* Annals of PDE **9** (2023), Art. 20, 53pp. DOI: 10.1007/s40818-023-00163-8, arXiv: 2207.08431

[39] A. Blumenthal, M. Coti Zelati & R.S. Gvalani, *Exponential mixing for random dynamical systems and an example of Pierrehumbert,* Annals of Probability **51** (2023), 1559–1601. DOI: 10.1214/23-AOP1627, arXiv: 2204.13651

[38] M. Coti Zelati & T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows,* Journal of the London Mathematical Society **108** (2023), 1358-1392. DOI: 10.1112/jlms.12782, arXiv: 2108.11192

[37] J. Bedrossian, R. Bianchini, M. Coti Zelati & M. Dolce, *Nonlinear inviscid damping and shear-buoyancy instability in the two-dimensional Boussinesq equations,* Communications on Pure and Applied Mathematics **76** (2023), 3685–3768. DOI: 10.1002/cpa.22123, arXiv: 2103.13713

[36] M. Coti Zelati, M. Dolce, Y. Feng & A.L. Mazzucato, *Global existence for the two-dimensional Kuramoto-Sivashinsky equation with a shear flow,* Journal of Evolution Equations **21** (2021), 5079–5099. DOI: 10.1007/s00028-021-00752-9, arXiv: 2103.02971

[35] M. Colombo, M. Coti Zelati & K. Widmayer, *Mixing and diffusion for rough shear flows,* Ars Inveniendi Analytica (2021), Paper No. 2, 21pp. DOI: 10.15781/83fc-j334, arXiv: 2009.12268

[34] M. Coti Zelati & G.A. Pavliotis, *Homogenization and hypocoercivity for Fokker-Planck equations driven by weakly compressible shear flows,* IMA Journal of Applied Mathematics **85** (2020), 951–979. DOI: 10.1093/imamat/hxaa035, arXiv: 2008.11710

[33] M. Coti Zelati, T.M. Elgindi & K. Widmayer, *Stationary structures near the Kolmogorov and Poiseuille flows in the 2d Euler equations,* Archive for Rational Mechanics and Analysis **247** (2023), Paper No. 12. DOI: 10.1007/s00205-023-01842-3, arXiv: 2007.11547

[32] R. Bianchini, M. Coti Zelati & M. Dolce, *Linear inviscid damping for shear flows near Couette in the 2D stably stratified regime,* Indiana University Mathematics Journal **71** (2022), 1467–1504. DOI: 10.1512/iumj.2022.71.9040, arXiv: 2005.09058

[31] M. Coti Zelati & M. Hairer, *A noise-induced transition in the Lorenz system,* Communications in Mathematical Physics **383** (2021), 2243–2274. DOI: 10.1007/s00220-021-04000-6, arXiv: 2004.12815

[30] M. Coti Zelati & T.D. Drivas, *A stochastic approach to enhanced diffusion,* Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834. DOI: 10.2422/2036-2145.201911_013, arXiv: 1911.09995

[29] M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit,* Journal of Functional Analysis **279** (2020), 108562. DOI: 10.1016/j.jfa.2020.108562, arXiv: 1909.01310

[28] M. Coti Zelati & M. Dolce, *Separation of time-scales in drift-diffusion equations on* $\mathbb{R}^2$*,* Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75. DOI: 10.1016/j.matpur.2020.08.001, arXiv: 1907.04012

[27] J. Bedrossian, M. Coti Zelati, S. Punshon-Smith & F. Weber, *Sufficient conditions for dual cascade flux laws in the stochastic 2d Navier-Stokes equations,* Archive for Rational Mechanics and Analysis **237** (2020), 103–145. DOI: 10.1007/s00205-020-01503-9, arXiv: 1905.03299

[26] M. Coti Zelati, T.M. Elgindi & K. Widmayer, *Enhanced dissipation in the Navier-Stokes equations near the Poiseuille flow,* Communications in Mathematical Physics **378** (2020), 987–1010. DOI: 10.1007/s00220-020-03814-0, arXiv: 1901.01571

[25] M. Coti Zelati, M.G. Delgadino & T.M. Elgindi, *On the relation between enhanced dissipation time-scales and mixing rates,* Communications on Pure and Applied Mathematics **73** (2020), 1205–1244. DOI: 10.1002/cpa.21831, arXiv: 1806.03258

[24] J. Bedrossian, M. Coti Zelati, S. Punshon-Smith & F. Weber, *A sufficient condition for the Kolmogorov 4/5 law for stationary martingale solutions to the 3D Navier-Stokes equations,* Communications in Mathematical Physics **367** (2019), 1045–1075. DOI: 10.1007/s00220-019-03396-6, arXiv: 1803.09695

[23] M. Coti Zelati, N. Glatt-Holtz & K. Trivisa, *Invariant measures for the stochastic one-dimensional compressible Navier-Stokes equations,* Applied Mathematics and Optimization **83** (2021), 1487–1522. DOI: 10.1007/s00245-019-09594-x, arXiv: 1802.04000

[22] M. Coti Zelati & C. Zillinger, *On degenerate circular and shear flows: the point vortex and power law circular flows,* Communications in Partial Differential Equations **44** (2019), 110–155. DOI: 10.1080/03605302.2018.1542436, arXiv: 1801.07371

[21] J. Bedrossian, M. Coti Zelati & V. Vicol, *Vortex axisymmetrization, inviscid damping, and vorticity depletion in the linearized 2D Euler equations,* Annals of PDE **5** (2019), Art. 4, 192pp. DOI: 10.1007/s40818-019-0061-8, arXiv: 1711.03668

[20] M. Coti Zelati & P. Kalita, *Smooth attractors for weak solutions of the SQG equation with critical dissipation,* Discrete and Continuous Dynamical Systems - Series B **22** (2017), 1857–1873. DOI: 10.3934/dcdsb.2017110, arXiv: 1512.08418

[19] M. Coti Zelati, *Long time behavior and critical limit of subcritical SQG equations in scale-invariant Sobolev spaces,* Journal of Nonlinear Science **28** (2018), 305–335. DOI: 10.1007/s00332-017-9409-y, arXiv: 1512.00497

[18] J. Bedrossian & M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows,* Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204. DOI: 10.1007/s00205-017-1099-y, arXiv: 1510.08098

[17] P. Constantin, M. Coti Zelati & V. Vicol, *Uniformly attracting limit sets for the critically dissipative SQG equation,* Nonlinearity **29** (2016), 298–318. DOI: 10.1088/0951-7715/29/2/298, arXiv: 1506.06375

[16] J. Bedrossian, M. Coti Zelati & N. Glatt-Holtz, *Invariant measures for passive scalars in the small noise inviscid limit,* Communications in Mathematical Physics **348** (2016), 101–127. DOI: 10.1007/s00220-016-2758-9, arXiv: 1505.07356

[15] M. Coti Zelati & V. Vicol, *On the global regularity for the supercritical SQG equation,* Indiana University Mathematics Journal **65** (2016), 535–552. DOI: 10.1512/iumj.2016.65.5807, arXiv: 1410.3186

[14] M. Coti Zelati & C.G. Gal, *Singular limits of Voigt models in fluid dynamics,* Journal of Mathematical Fluid Mechanics **17** (2015), 233–259. DOI: 10.1007/s00021-015-0201-1, arXiv: 1408.3497

[13] M. Coti Zelati & P. Kalita, *Minimality properties of set-valued processes and their pullback attractors,* SIAM Journal on Mathematical Analysis **47** (2015), 1530–1561. DOI: 10.1137/140978995, arXiv: 1407.6135

[12] M. Coti Zelati, A. Huang, I. Kukavica, R. Temam & M. Ziane, *The primitive equations of the atmosphere in presence of vapour saturation,* Nonlinearity **28** (2015), 625–668. DOI: 10.1088/0951-7715/28/3/625, arXiv: 1406.3165

[11] A. Bousquet, M. Coti Zelati & R. Temam, *Phase transition models in atmospheric dynamics,* Milan Journal of Mathematics **82** (2014), 99–128. DOI: 10.1007/s00032-014-0213-y

[10] M. Coti Zelati, M. Frémond, R. Temam & J. Tribbia, *The equations of the atmosphere with humidity and saturation: uniqueness and physical bounds,* Physica D **264** (2013), 49–65. DOI: 10.1016/j.physd.2013.08.007

[9] M. Coti Zelati, *Remarks on the approximation of the Navier-Stokes equations via the implicit Euler scheme,* Communications on Pure and Applied Analysis **12** (2013), 2829–2838. DOI: 10.3934/cpaa.2013.12.2829

[8] M. Coti Zelati, *On the theory of global attractors and Lyapunov functionals,* Set-Valued and Variational Analysis **21** (2013), 127–149. DOI: 10.1007/s11228-012-0215-2

[7] M. Coti Zelati, F. Dell'Oro & V. Pata, *Energy decay of type III linear thermoelastic plates with memory,* Journal of Mathematical Analysis and Applications **401** (2013), 357–366. DOI: 10.1016/j.jmaa.2012.12.031

[6] M. Coti Zelati & F. Tone, *Multivalued attractors and their approximation: applications to the Navier-Stokes equations,* Numerische Mathematik **122** (2012), 421–441. DOI: 10.1007/s00211-012-0463-y, arXiv: 1111.4368

[5] M. Coti Zelati & R. Temam, *The atmospheric equation of water vapor with saturation,* Bollettino dell'Unione Matematica Italiana **5** (2012), 309–336.

[4] M. Conti & M. Coti Zelati, *Attractors for the Cahn–Hilliard equation with memory in 2D,* Nonlinear Analysis **72** (2010), 1668–1682. DOI: 10.1016/j.na.2009.09.006

[3] M. Coti Zelati, C. Giorgi & V. Pata, *Steady states of the hinged extensible beam with external load,* Mathematical Models and Methods in Applied Sciences **20** (2010), 43–58. DOI: 10.1142/S0218202510004143

[2] M. Coti Zelati, V. Pata & R. Quintanilla, *Regular global attractors of type III thermoelastic extensible beams,* Chinese Annals of Mathematics **31** (2010), 619–630. DOI: 10.1007/s11401-010-0605-4

[1] M. Coti Zelati, *Global and exponential attractors for the singularly perturbed extensible beam,* Discrete and Continuous Dynamical Systems **25** (2009), 1041–1060. DOI: 10.3934/dcds.2009.25.1041